

CDA Homework 5

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Answer 1:

The architecture uses a branch target buffer (BTB) to predict the target address for branch instructions. The BTB contains the address of the last branch instruction executed and its target. When a new branch instruction is fetched, the BTB is checked to see if there is a matching entry based on the branch address. If there is a match, the predicted target from the BTB is used. This allows the processor to begin executing along the predicted path while the branch condition is evaluated. The 16-bit compressed instructions can encode smaller versions of branch instructions that index into the BTB in the same way.

Answer 2:

- ID is on the first and MEM on second. The critical path delay will be 250.
- The percentage of instructions using “immediate generation unit” is $20\% + 20\% + 15\% = 55\%$
- 100% of the instructions use instruction memory.

Answer 3:

- The R-type instruction will fail.
- The ALU and R-type instructions will fail.

Answer 4:

1	2	3	4	5	6	7	8	9	10	11
	ID	EX	ME	WB						
	IF	ID	EX	ME	WB					
		IF	ID	EX	ME	WB				
				IF	ID	EX	ME	WB		
					IF	ID	EX	ME	WB	
						IF	ID	EX	ME	WB

Answer 5:

- Minimum number of instructions: 100ns
- Single cycle time/ multiple time = 1.5; $100N/(200N+50N) = 1.5$
- At a steady state, we have 1 new instruction completed every 1 cycle = every 50 ns. Previously (with the single cycle implementation) this was 100ns. Hence, the maximum speedup is $100 \text{ ns} / 50 \text{ ns} = 2x$.

Answer 6:

- The $n+1$ bit counter is used to detect empty and full conditions. The empty output is asserted when the counter is 0, and the full output is asserted when the counter hits $n+1$.
- The two comparing circuits are used to compare the read and write pointers to detect empty and full conditions. This avoids having to propagate the carry chain through the entire counter.
- The data signals cross the clock domains. The FIFO design uses gray code pointers and registers to synchronize signals and avoid metastability issues.
- For multi-signal CDCs, separate FIFOs can be used for each signal. The FIFOs will synchronize and align the data from each clock domain. The output will have all signals from one clock aligned to the destination clock domain.